

ABID HASAN

Bachelor of Science (BSc) in Electrical and Electronic Engineering

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Links: [LinkedIn](#) | [ResearchGate](#) | [Google Scholar](#) | [ORCID](#)

EDUCATION

- **Bachelor of Science (BSc) in Electrical and Electronic Engineering** | Rajshahi University of Engineering & Technology (RUET) | 2022 - Present
Current CGPA: 3.56/4.00 (Till 7th Semester)

THESIS WORK

Thesis Title: Automated Sleep Stage Classification in Sleep Disorders Using Multimodal Wearable Signals

Role: Bachelor's Thesis Student/ Student Researcher

Supervisor: Dr. Mohammad Abdul Motin | **Website:** <https://sites.google.com/view/drmamotin>

Institution: Rajshahi University of Engineering & Technology, Rajshahi-6204, Bangladesh

Thesis Overview: Developed M-SleepNet, a multimodal deep learning framework for automated sleep-stage classification in individuals with sleep disorders using PPG and accelerometry (ACC) signals with HRV derived from raw PPG. The framework integrates modality-specific residual convolutional encoders, channel and cross-modal attention, gated fusion, temporal convolutional networks (TCN), and bidirectional GRU with multi-head self-attention to capture multimodal and temporal dependencies. The proposed framework was developed and evaluated using the DREAMT wearable sleep dataset.

RESEARCH INTERESTS

- Biomedical Signal Processing
- Machine Learning for Healthcare
- Wearable Health Monitoring Systems
- Internet of Medical Things (IoMT)
- Embedded AI and Edge Computing

RESEARCH EXPERIENCES

Research Assistant | Signal Processing and Machine Learning Lab, Dept. of EEE, RUET | July 2024 – Present

Lab Profile: <https://drmmamotin.vercel.app/lab/abid-hasan>

- Conducting research on biomedical signal processing and machine learning for edge IOT devices

Publications

- TinyArtifactNet: A Resource-Efficient Deep Learning Model for On-Device Motion Artifact Detection in PPG
IEEE Sensors Journal, 2026
DOI: [10.1109/JSEN.2026.3692462](https://doi.org/10.1109/JSEN.2026.3692462)
– Proposed a lightweight deep learning framework for real-time PPG motion artifact detection on embedded/wearable devices.
Role: Data preprocessing, Model development, experimentation, and manuscript preparation.

- A Multi-modal Wearable Dataset for Cognitive Attention and Task-based Stress Analysis
Data in Brief, 2025
DOI: [10.1016/j.dib.2025.112295](https://doi.org/10.1016/j.dib.2025.112295)
– Developed and documented a multimodal physiological dataset for cognitive attention and stress research.
Role: Data curation and original draft writing

SKILLS

- **Programming & Development:** Python, MATLAB, C/C++
- **Machine Learning & AI:** Deep Learning, TensorFlow / Keras, CNN, LSTM, Transformer Model
- **Biomedical Signal Processing:** Data Acquisition, Feature Extraction & Data Preprocessing
- **Tools & Software:** Git & GitHub, Jupyter Notebook, Kaggle, VS Code, MATLAB, ArduinoIDE, STM32CubeIDE, Fusion 365, AutoCAD, Multisim
- **Research & Analysis:** Scientific Writing, Data Analysis, Statistical Analysis
- **Embedded Systems:** STM32 Development, Real-Time Data Analysis, Sensor Interfacing & Integration, Circuit Design
- **Leadership & Management:** Team Coordination, Project Management, Research Collaboration

INDUSTRIAL ATTACHMENT

- **Industrial Trainee** at Civil Aviation Authority of Bangladesh (CAAB) – March 2025
- **Industrial Trainee** at THiNK – March 2025

CERTIFICATIONS

- Supervised Machine Learning: Regression and Classification – Coursera
- Introduction to Programming with MATLAB – Coursera

ACADEMIC PROJECTS & WORKSHOPS

- **BIOMETRIC BEACON:** Real-time embedded health monitoring system for biometric data acquisition and analysis, presented at EEE-3100: Electronic Shop Practice event
- **Capstone Project I (EEE-4100):** TinyMA-Net: Lightweight deep learning model for real-time motion artifact detection in PPG signals on embedded systems

VOLUNTEERING & COMMUNITY ENGAGEMENT

- Volunteer, ASSR “AURORA: National Science Festival 2023” – Supported event coordination and execution
- Event Organizer, TIB RUET DC IV 2022, organized by RUET Debating Club
- Served as Promotion & Branding Secretary of the Astronomy & Science Society of RUET
- Served as Community Coordinator of the RUET Esports Community

COMPETITIONS

- Participated in the IEEE Signal Processing Cup – “Deepfake Face Detection In The Wild” as a member of Team “**Pseudo Spectrum**” (Team ID: 28267)
- Secured the **2nd Runner-up** position in the Fastest Line Follower Contest at Expeditious 1.0, hosted by the Department of Electrical and Electronic Engineering, RUET

LANGUAGE SKILLS

- English
- Bangla